BTP Reading Notes

Perceptual / Deep Hashing — Core Reading List (6 papers)

FOUNDATIONAL ATTACK

Learning to Break Deep Perceptual Hashing: The Use Case NeuralHash

Struppek, Hintersdorf, Neider, Kersting
ACM FAccT 2022 · doi.org/10.1145/3531146.3533073

Core contribution: First thorough empirical break of NeuralHash — gradient-based and standard image edits both induce/avoid hash collisions. The foundational 'deep hashing is fragile' result everything else responds to.

Pass 1 — Skim Notes

- First comprehensive empirical security analysis of NeuralHash specifically - prior work only had isolated PoC collisions, no systematic study.
- Perceptual hash pipeline: embedding network (modified MobileNetV3, 1.8M params, self-supervised contrastive training) extracts a 128-dim feature vector -> LSH step multiplies it by a hashing matrix B (128x96, i.e. 96 random hyperplanes) -> each bit = which side of a hyperplane the feature vector falls on -> 96-bit binary hash.
- Perceptual hash vs cryptographic hash: no avalanche effect wanted - similar images should get similar/close hashes (measured via normalized Hamming distance), not wildly different ones.
- Four adversarial settings tested: (1) hash collision attacks, (2) gradient-based evasion attacks, (3) gradient-free evasion attacks (standard image transforms), (4) hash information extraction (leakage).

Pass 2 — Close-Read Notes

- Setup: white-box - full access to the model and its gradients.
- Collision attack: pick a target hash from a (surrogate) hash database, directly optimize the input image x with gradient descent (Adam) using a Hinge loss that pushes the pre-binarization output y = B*M(x) to match the target hash's sign pattern, plus an SSIM penalty to keep the image visually close to the original. ~90% success rate, ~1190 steps on average, image barely visibly changed (SSIM ~0.96).
- Gradient evasion attack: same idea reversed - maximize Hamming distance from the *original* hash instead of matching a target, via a negative-MSE loss (with a sigmoid standing in for the non-differentiable binarization step) + SSIM penalty. Three variants: Standard (perturb whole image), Edges-Only (restrict to Canny edge pixels), Few-Pixels (greedily add pixels one at a time until a bit flips). Success rates 100% / 99.95% / 98.21%; Standard attack flips a bit in ~5 steps on average - trivially easy.
- Gradient-free evasion: black-box, no gradients needed - just apply ordinary image-editor transforms (rotate, translate, crop, flip, brightness/contrast, JPEG compression) and measure the resulting Hamming distance. Rotation/cropping/flipping change a large fraction of bits (e.g. vertical flip changes ~37% of bits); compression and HSV changes are comparatively robust.
- Hash leakage: train a simple classifier that takes only the 96-bit hash as input and predicts the image's class - gets 4.34% top-1 on 1000 ImageNet classes vs 0.1% random chance (higher again on coarser 85-category grouping), proving the hash itself encodes recoverable content information.
- Conclusion: NeuralHash (and deep perceptual hashing broadly) is not robust - trivially broken by both gradient-based and gradient-free manipulation, and its hashes leak content info, undermining the privacy case for client-side scanning. Authors call for it not to be deployed as-is.
- Mitigations discussed (server-side re-hashing, restricting gradient access, adversarial training) each trade away part of what client-side scanning was supposed to provide (privacy, feasibility) or don't fully close the gap - no

clean fix.

CERTIFIED DEFENSE

CertPHash: Towards Certified Perceptual Hashing via Robust Training

Yang, Liu, Brix, Zhang, Cao
USENIX Security 2025 · usenix.org/conference/usenixsecurity25/presentation/yang-yuchen

Core contribution: First certified PHash system: bound-propagation training with anti-evasion, anti-collision, and functionality loss terms, giving provable (not just empirical) robustness guarantees. Core paper for your BTP.

Pass 1 — Skim Notes

- Three training objectives: anti-evasion, anti-collision, functionality loss.
- Proven highly resilient against actual empirical gradient-based attacks.
- Evasion = evade content filters. Collision = make a benign image match an illicit one.
- Perturbation = tiny calculated deviation from the original hash, e.g. via projected gradient descent, then projected back into the allowed perturbation boundary.
- Trains existing PHash models with a new loss function (for non-neural hashes, uses a surrogate deep network).

Pass 2 — Close-Read Notes

- How does the attacker know the target hash? Either (a) full access to the PHash model, or (b) the illicit-hash database is stored client-side (e.g. Apple's NeuralHash, for privacy) or the blocklist/allowlist is publicly accessible - so the attacker can obtain target hashes either way.
- Uses Forward Verification and Backward Verification.

Gaps Identified

- Training bottleneck: certified/verification training is slow.
- Suggested future work: design 'verification-friendly' neural networks (e.g. pruning unstable neurons) to speed up the verifier.

PROVENANCE / ARCHITECTURE

Provenance Detection for AI-Generated Images: Combining Perceptual Hashing, Homomorphic Encryption, and AI Detection Models (DinoHash)

Singhi, Yadav, Gupta, Ebrahimi, Hassanizadeh, et al.
arXiv 2503.11195 (Mar 2025) · arxiv.org/abs/2503.11195

Core contribution: Builds a hash on frozen/fine-tuned DINOv2 features + PCA whitening + LSH (96-bit), paired with MP-FHE for private querying. Aimed at tracing AI-generated images through screenshots/re-uploads rather than CSAM-style matching. Core paper for your BTP.

Pass 1 — Skim Notes

- Framework for robust perceptual image hashing for provenance detection of AI-generated images.
- Uses deep neural networks to extract salient features from an image, used to calculate a compact binary hash.

- Goal: images can be attributed/copyrighted openly and transparently.

Pass 2 — Close-Read Notes

- Pipeline: deep feature extractor -> binarization via LSH (Locality Sensitive Hashing).
- Each hash bit = which side of a hyperplane the feature vector lies on; output is a 96-bit binary hash.
- Feature extractor = DINOv2, 768-dim output, naturally resilient to crop/blur/color-shifting.
- PCA whitening = removes correlation between feature dimensions.
- To prevent hash collapse: a student-distillation pipeline is set up.

BACKDOOR / POISONING

DarkHash: A Data-Free Backdoor Attack Against Deep Hashing

Zhou, Deng, Song, Zhang, Wan, Hu, Li, Zhang, Yao
IEEE TIFS 2025 / arXiv 2510.08094 · arxiv.org/abs/2510.08094

Core contribution: First data-free backdoor: embeds a backdoor into a deployed deep hashing model without access to its original training data, via a shadow-model + topological-alignment loss, using only a surrogate dataset. Survives mainstream backdoor defenses. Core paper for your BTP.

Pass 1 — Skim Notes

- Dual-Semantic Guidance: fine-tunes with surrogate data while keeping primary feature extraction fixed, to implant the backdoor.
- Using a top-k loss function makes the attack stronger.
- Model behaves normally on benign inputs, but outputs the attacker-designed (malicious) hash when the trigger is present.

Pass 2 — Close-Read Notes

- Loss function = L_ben + L_bac + L_tpa (benign + backdoor + topological-alignment terms).
- Backdoor defenses tested against: model diagnosis (analyze trained network architecture) and input diagnosis (scan inputs, block trigger patterns).

Gaps Identified

- No defense models exist to protect deep neural hash models.

SURVEY / BACKGROUND

Perceptual Hashing for Image Authentication: A Survey

Du, Ho, Cong
Signal Processing: Image Communication, 2020 · doi.org/10.1016/j.image.2020.115713

Core contribution: Pre-deep-learning-era survey of classical perceptual hashing for authentication. Good for framing the 'before neural hashing' baseline in your intro/related work.

Pass 1 — Skim Notes

- Cryptographic hash vs perceptual hash: crypto hash is highly sensitive (tiny change = totally different hash) -> used for integrity checking. Perceptual hash uses visual/content characteristics, resilient to small changes, no key, resistant to collisions - good for similarity matching.

- Hash generation draws on freq, edge, corner, texture, color, salient regions, etc.
- Transform Domain methods - Radon/DCT/Fourier transforms, robust to low-frequency filtering & compression; weakness is attack-specific robustness and poor local-tamper info.
- Local feature point methods - SIFT/salient regions/shape content, robust to rotation/crop/geometric change/local tamper; but unrelated images may share common features.
- Dimension reduction methods - PCA/LLE/SVD/NMF/compressive sensing, reduces storage but may discard information; sensitive to geometric changes.
- Statistical feature methods - mean/var/entropy/histograms, loses spatial info; different images can have similar global statistics.
- Learning-based methods - CNN/Transformers, may fail on unseen attacks.
- General note: increasing the matching threshold -> more false positives.

Gaps Identified

- Many hash generation methods are designed for/robust to only a few attack types - narrow, attack-specific robustness rather than general robustness.

SURVEY / BACKGROUND

A Survey of Perceptual Hashing for Multimedia

(2025)
2025 · researchgate.net/publication/390471181

Core contribution: More recent survey of the broader perceptual-hashing-for-multimedia space (not hashing-security specific) — worth a skim to check it doesn't already cover your gap before you commit to the survey angle.

Pass 1 — Skim Notes

- Image hashing coverage is largely the same as the 2020/'21 survey - new material is (1) neural network model hashing and (2) video hashing.
- Model Hashing (NN models): generates a compact fingerprint from a model's parameters, architecture, and learned behaviour - framed as an IP-protection/model-fingerprinting problem.
- Binary hash approach - extract & locally normalize input params, extract features via convolutional layers, compress into the hash.
- Graph-based approach - convert the architecture itself into a graph and run a GNN over it to create the hash.
- Video Hashing: harder than image hashing - high computation, difficult keyframe selection, complex motion, more audio to account for; needs both spatial info (per-frame) and temporal info (motion, frame order, scene transitions).
- Classical spatial methods select keyframes and apply image-hashing to them - not good for motion, quality depends heavily on keyframe selection.
- Spatial-temporal & learning-based video methods combine appearance and motion using domain-transform methods.
- Twinned CNNs: deep metric learning, conditional restricted Boltzmann machines, two-branch networks that separately model temporal notion and spatial frame structure.
- Unsupervised approaches: transformers, visual-completion tasks, 3D self-attention, multiscale feature fusion, pseudo-labels, dual constraints.